C. Dynamic Light Scattering

1. Overview

This section does not cover the theory in the same level of detail as for mass photometry, since DLS is a well-established technique and many well-written sources are available. These include standard textbooks [21–23] as well as seminal publications [24–27].

Dynamic Light Scattering provides information about particle motion by measuring how fast the scattered light intensity loses correlation in time. The decay is governed by the scattering wavevector

$$q = \frac{4\pi n}{\lambda} \sin\frac{\theta}{2}, \tag{31}$$

where n is the refractive index of the medium, λ is the incident light wavelength in vacuum, and θ is the scattering angle. This wavevector defines the spatial scale to which the experiment is sensitive. Correlation is lost once Brownian motion displaces a particle by a distance of order

$$|\Delta\mathbf{r}| \sim \frac{1}{q}, \tag{32}$$

corresponding to a phase change of order one in the scattered light. For diffusive motion in three dimensions,

$$\langle \Delta r^2(\tau)\rangle = 6D\tau, \tag{33}$$

where the angle brackets denote averaging over time. Thus, the characteristic time τ_c required to lose correlation through particle motion can be estimated via

$$6D\tau_c \sim \frac{1}{q^2}, \tag{34}$$

which leads to

$$\tau_c \sim (Dq^2)^{-1}. \tag{35}$$

As a result, larger particles (with smaller diffusion coefficients) decorrelate more slowly than smaller ones.

Autocorrelation is a way of asking a simple question: how similar is the scattering pattern at time $t+\tau$ to the pattern at time t? If the particle positions change only slightly during a delay time τ, the scattering pattern at time $t+\tau$ closely resembles that at time t, and the correlation remains high. Once particle motion significantly rearranges the interference pattern, the correlation decays. The autocorrelation function therefore measures how fast the scattering pattern "forgets" its initial configuration. Practically, calculating the correlation for the whole range of τ which is available to the instrument is a way to find the τ_c at which signal drops half. That is exactly the time in Eq. (35).

The scattered electric field carries a phase factor

$$E(t) \propto \exp\big(i\mathbf{q}\cdot\mathbf{r}(t)\big). \tag{36}$$

The field autocorrelation therefore depends on the accumulated phase difference

$$\Delta\phi(\tau) = \mathbf{q}\cdot\big[\mathbf{r}(t+\tau) - \mathbf{r}(t)\big]. \tag{37}$$

For Brownian motion, particle displacements are Gaussian distributed, and the phase variance is

$$\langle \Delta\phi^2(\tau) \rangle = q^2 \langle \Delta r^2(\tau) \rangle. \tag{38}$$

The phase variance therefore grows linearly with delay time. Using Eq. (33), for a Gaussian random phase the field autocorrelation becomes

$$g_1(\tau) = \langle e^{i\Delta\phi(\tau)} \rangle = \exp\big(-\tfrac{1}{2}\langle \Delta\phi^2(\tau)\rangle\big) \simeq \exp(-Dq^2\tau), \tag{39}$$

up to numerical factors of order unity set by the isotropic average.

As a result, the field autocorrelation decays exponentially,

$$g_1(\tau) = \langle E(t)\, E^*(t+\tau) \rangle = \exp(-Dq^2\tau), \tag{40}$$

reflecting the continuous loss of phase coherence as particles diffuse over the optically defined length scale $1/q$.

To reiterate, a DLS instrument compares the phases of the incident and scattered light waves. When particle motion shifts the phase of the scattered wave by roughly half a wavelength relative to the incident wave, the interference signal is lost and the correlation decays. Information about diffusion is therefore obtained from how fast this decay occurs: because the phase of the scattered light is driven by particle diffusion, faster-moving particles produce a faster exponential decay. Long measurement times, $t_{\text{meas}} \gg \tau_c$, are required only to improve the signal-to-noise ratio and do not affect the characteristic decay time itself. Consequently, all DLS data analysis ultimately reduces to analyzing the decay of an exponential correlation function.

2. *Heterodyne and homodyne detection*

There are two principal detection modes in dynamic light scattering: heterodyne and homodyne detection. They differ in their dynamic range and in how experimental noise propagates into the measured signal.

In heterodyne detection, the detected intensity results from interference between the scattered field $E_s(t)$ and a stable reference field $E_r(t)$. The fluctuating component (the last term on the right) is linear in the scattered field

$$I(t) = |E_r + E_s|^2 = |E_r|^2 + |E_s|^2 + 2\,\mathrm{Re}\{E_r^* E_s\}. \tag{41}$$

The measured intensity autocorrelation is therefore proportional to the field correlation function,

$$g_2(\tau) - 1 \propto g_1(\tau). \tag{42}$$

Because the observable depends linearly on the field fluctuations, noise propagates approximately linearly as well, and the resulting residuals can often be treated as additive and roughly homoscedastic.

In homodyne detection no external reference field is present and the detector measures only the scattered intensity,

$$I(t) = |E_s(t)|^2. \tag{43}$$

The intensity correlation function is related to the field correlation through the Siegert relation,

$$g_2(\tau) - 1 = \beta\, |g_1(\tau)|^2, \tag{44}$$

where β is the instrumental coherence factor. Recovering the field correlation function therefore requires

$$g_1(\tau) = \sqrt{\frac{g_2(\tau) - 1}{\beta}}. \tag{45}$$

Because the observable depends quadratically on the field amplitude, fluctuations enter multiplicatively and the square-root transformation introduces nonlinear noise propagation. The resulting noise is signal dependent and generally heteroscedastic. Since the analysis of dynamic processes is performed on the field correlation function $g_1(\tau)$, this transformation directly determines the statistical structure of the data used for further analysis.

The two detection schemes therefore differ in both signal structure and noise propagation:

Detection mode	g_1 reconstruction	Noise propagation
Heterodyne	$g_1(\tau) \propto g_2(\tau) - 1$	linear, approximately homoscedastic
Homodyne	$g_1(\tau) = \sqrt{\frac{g_2(\tau) - 1}{\beta}}$	nonlinear, heteroscedastic

TABLE IV. Comparison of heterodyne and homodyne detection in dynamic light scattering.

Thus, heterodyne detection provides a linear measurement of the field correlation function, whereas homodyne detection measures its square through the intensity signal, leading to fundamentally different noise characteristics in the reconstructed $g_1(\tau)$. Unfortunately most commercial instruments operate in the homodyne mode because it is cheaper to implement, which limits the effective dynamic range.

3. Polydispersity and flexibility. Data inversion.

The above description implicitly assumes a monodisperse sample with rigid particles, a situation that rarely occurs in practice even for reference samples. Real samples are typically polydisperse and often flexible (e.g., polymers or fractal aggregates). The field correlation function $g_1(\tau)$ reflects only how the scattered field decorrelates in time and does not specify the physical origin of this motion. Consequently, both flexibility and polydispersity contribute to the observed shape of $g_1(\tau)$.

For flexible structures, such as fractal aggregates [28] and semiflexible polymers including amyloid filaments, actin filaments, and DNA, the translation of the center of mass may not coincide with the motion of individual molecular segments relative to it. For example, when the contour length of a filament exceeds its persistence length ($L \gg \ell_p$), thermally driven bending of different segments introduces internal relaxation modes partially decoupled from translational diffusion. These internal fluctuations contribute additional decorrelation channels to $g_1(\tau)$ [29–31]. In contrast, polydispersity leads to particles moving with different diffusion coefficients, producing a distribution of characteristic decay times τ. For fractal aggregates, however, the effects of polydispersity typically dominate the overall correlation decay [28].

Mathematically, for a polydisperse and flexible system, the field correlation function can be written as a superposition of relaxation modes arising from translational diffusion and internal motions,

$$g_1(\tau) = \sum_i a_i e^{-D_i q^2 \tau} + \sum_j b_j e^{-\lambda_j \tau}, \tag{46}$$

where the first term describes decorrelation due to translational diffusion with diffusion coefficients D_i (polydispersity), and the second term represents internal relaxation modes with rates λ_j arising from flexibility of the scattering objects. The coefficients a_i and b_j represent the corresponding distribution weights. In reality, however, these coefficients are coupled and the sum in Eq. 46 is much more complex [29–31].

Nevertheless, as can be seen from the forward model (Eq. 46), if the distribution parameters a_i, b_j and the diffusion coefficients D_i and internal relaxation rates λ_j are known, the correlation function $g_1(\tau)$ can be constructed directly. In practice, however, the experiment measures $g_1(\tau)$ and the goal is to recover the underlying parameters a_i, b_j, D_i, and λ_j. This constitutes an inverse problem. The problem is ill-posed in the sense that many different combinations of these parameters can reproduce nearly identical correlation functions, while experimental noise can strongly distort the recovered solution.

For rigid polydisperse particles, Eq. (46) reduces to a superposition of translational diffusion modes,

$$g_1(\tau) = \sum_i a_i\, e^{-\Gamma_i \tau}, \tag{47}$$

where $\Gamma_i = D_i q^2$ are the decay rates associated with different particle sizes, and the coefficients a_i represent the relative contribution of species i. When normalized such that $\sum_i a_i = 1$, the set $\{a_i\}$ defines the distribution of interest, where normalized a_i correspond to intensity-weighted probabilities. In the following example, the discrete index i is identified with aggregation number N.

To illustrate the ill-posed nature of the inversion problem, the power-law distribution in Eq. (6) is considered,

$$N = 1, 2, \ldots, 100, \qquad a(N) \propto N^{-1.5}, \qquad a(N) = \frac{N^{-1.5}}{\sum_{N=1}^{100} N^{-1.5}}, \tag{48}$$

where N is the aggregation number, with $N = 1$ corresponding to the monomer.

For each N, the diffusion coefficient is computed using the Stokes–Einstein relation,

$$D(N) = \frac{k_B T}{6\pi\eta\, R(N)}, \tag{49}$$

where $R(N)$ is interpreted as an effective radius of an N-mer. The corresponding decay rates are

$$\Gamma(N) = D(N)\, q^2, \qquad q = \frac{4\pi n}{\lambda} \sin\frac{\theta}{2}. \tag{50}$$

> In the simulations, the monomer radius was taken as $R_0 = 3$ nm. Here, $R(N)$ is treated as an effective radius derived from aggregate mass. This approximation is not exact for small N (e.g. a dimer has mass $2m$ but a radius smaller than $2R_0$), but becomes increasingly reasonable for larger aggregates. The instrumental parameters were $\lambda = 633$ nm, $n = 1.33$, and $\theta = 173°$, corresponding to the laser wavelength in vacuum, refractive index of the medium, and scattering angle, respectively.

Two sets of curves are plotted in Figure 7: (i) normalized correlation functions $\frac{g_1(\tau)}{g_1(0)}$ for individual aggregate sizes N (colored from cyan to black), and (ii) the intensity-weighted ensemble correlation function together with a single-exponential reference constructed from the intensity-weighted average decay rate of the ensemble.

The ensemble correlation function is

$$g_{1,\mathrm{ens}}(\tau) = \sum_{N=1}^{100} a(N)\, e^{-\Gamma(N)\tau}, \tag{51}$$

where $a(N)$ are the normalized intensity weights of aggregates with aggregation number N.

The corresponding intensity-weighted average decay rate is

$$\langle\Gamma\rangle = \sum_{N=1}^{100} a(N)\,\Gamma(N), \tag{52}$$

from which the single-exponential reference is constructed as

$$g_{1,\mathrm{avg}}(\tau) = \exp(-\langle\Gamma\rangle\tau)\,. \tag{53}$$

These constructions illustrate an important consequence of Jensen's inequality for DLS systems: averaging and exponentiation do not commute,

$$\left\langle e^{-x}\right\rangle \neq e^{-\langle x\rangle}. \tag{54}$$

Consequently, the ensemble-averaged autocorrelation function does not generally correspond to a single exponential constructed from the average relaxation time.

Figure 7 illustrates the fundamental difficulty of extracting size distributions from DLS autocorrelation data. Assuming a noise level comparable to the line thickness, only the first few decay modes (approximately five) produce experimentally distinguishable deviations in the time-domain representation, while higher-order modes progressively collapse into a smooth envelope. This *dynamic compression*, originating from the inverse scaling of diffusion coefficients with aggregate size, causes many exponential components to become effectively degenerate within experimental noise.

As a consequence, directly fitting a large sum of exponentials is ill-conditioned: many different combinations of amplitudes and decay rates produce nearly identical correlation curves. This strong parameter correlation is the main source of instability and noise sensitivity in multi-exponential fits and explains why cumulant analysis, which retains only the lowest moments of the distribution, is widely used in practice.

Figure 7 also shows that the ensemble-averaged autocorrelation (*dashed* line, Eq. (51)) does not coincide with a single exponential constructed from the mean relaxation time (*dotted* line, Eq. (53)), demonstrating that without explicit knowledge of polydispersity form, even the average particle size cannot be reliably inferred from a single-exponential fit.

To overcome the above difficulties, a range of methods has been developed [21–23], with two of the most widely used approaches being cumulant analysis (ISO 22412) and the Maximum Entropy Method (MEM).

4. Inversion methods

Before the cummulants and MEM methods are discussed, one model-free way to estimate deviations from single-exponential decay is through nondimensionalisation of the time axis. The correlation function is first fitted with $\exp(-\tau/\tau_c)$ and the time axis is rescaled as τ/τ_c, such that time is measured in units of the decay time τ_c rather than in seconds. If the experimental autocorrelation function is truly single-exponential, it will appear as a straight line on a log–linear plot, since $\log[\exp(x)] = x$. Any deviation from linearity therefore directly reflects departures from single-exponential behavior, as illustrated in Figure 8A. As seen in Figure 8B, the effects of polydispersity are to a large extent contained in the long-time tails of the correlation function.

In the cumulant approach, the field autocorrelation function is expanded around short delay times as a series in powers of τ. Writing the normalized field autocorrelation function as $g_1(\tau)$, its logarithm is expressed as

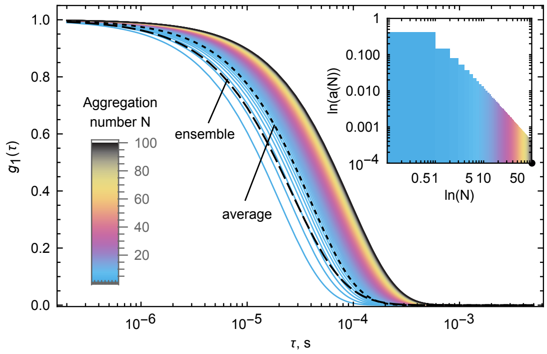

FIG. 7. Ill-posed nature of fitting DLS data for a polydisperse sample. Only the first few modes produce experimentally distinguishable deviations, while higher modes progressively merge due to dynamic compression. The *dashed* black line shows the weighted ensemble correlation function $g_{1,\mathrm{ens}}$ [Eq. (51)], while the *dotted* line represents the single-exponential decay $g_{1,\mathrm{avg}}$ constructed from the average relaxation time [Eq. (53)]. Insets: color bar indicating aggregation number (lower left); power-law coefficients $a(N)$ plotted versus N on a log–log scale [Eq. (48)] (upper right).

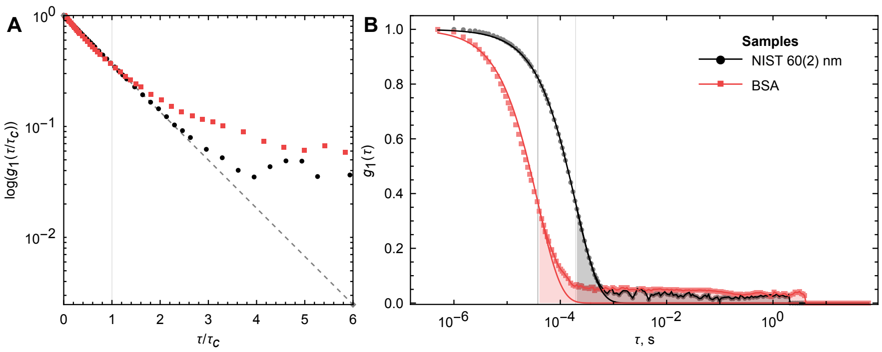

FIG. 8. Deviation from single-exponential decay in a polydisperse sample. A—First-order field autocorrelation function $g_1(\tau)$ plotted versus the dimensionless time τ/τ_c on a log–linear scale. Black symbols: NIST polystyrene size standard with very low polydispersity (60 ± 2 nm). Red symbols: BSA sample containing a mixture of monomers, dimers, trimers, and tetramers. Vertical lines indicate the characteristic relaxation times τ_c obtained from single-exponential fits of the form $\exp(-\tau/\tau_c)$. The dotted line corresponds to $\exp(-\tau/\tau_c)$ with $\tau_c = 1$. B—First-order autocorrelation function $g_1(\tau)$ shown on a linear–log scale as a function of real time. The shaded gray region highlights the deviation from single-exponential behavior due to polydispersity. Symbols represent experimental data points, and solid lines are single-exponential fits of the form $\exp(-\tau/\tau_c)$.

$$\ln g_1(\tau) = -\Gamma_1\tau + \frac{\Gamma_2}{2!}\tau^2 - \frac{\Gamma_3}{3!}\tau^3 + \cdots , \tag{55}$$

where Γ_1 is the first cumulant and $\Gamma_2, \Gamma_3, \ldots$ are higher-order cumulants. The first cumulant corresponds to the average decay rate,

$$\Gamma_1 = \langle\Gamma\rangle = D_z q^2, \tag{56}$$

from which the so-called Z-average diffusion coefficient D_z and the Z-average hydrodynamic size are obtained. The second cumulant quantifies the width of the decay-rate distribution and is commonly reported in normalized form as the polydispersity index (PDI),

$$\mathrm{PDI} = \frac{\Gamma_2}{\Gamma_1^2}. \tag{57}$$

Higher-order cumulants are increasingly sensitive to noise and are therefore rarely used in practice. As a result, ISO standard DLS analysis typically reports only the Z-average size and the polydispersity index (PDI), which provide a robust but limited description of polydispersity without attempting to recover the full size distribution. The term *Z-average* is historical rather than mathematical in origin and does not refer to a statistical z-score or a spatial coordinate, but simply denotes the intensity-weighted mean diffusion coefficient used in the cumulants formalism.

Maximum Entropy Method is based on the concept of Shannon entropy, which favors solutions containing the least additional structure beyond what is required by the data. When the flexibility contribution is negligible, the second term in Eq. (46) vanishes and the amplitudes a_i in the first term of Eq. (46) are determined by entropy maximization subject to the experimental constraints. This yields the least biased distribution compatible with the measured correlation function through the constraint in Eq. (58).

Up to this point, the discussion has been purely theoretical, implicitly assuming ideal data. Real experimental data are always noisy, so any practical implementation of the Maximum Entropy Method must account for noise. Traditionally, this is done using a χ^2 statistic [26]. If the experimental errors are approximately normally distributed the total χ^2 is of the order of the number of data points M in the measured correlation function $g_1(\tau)$ [22]. In this case, the simplest MEM formulation combines the Shannon entropy and the constraint,

$$g_1(\tau) = \sum_i a_i e^{-D_i q^2 \tau} \qquad S(\mathbf{a}) = -\sum_i a_i \ln a_i, \qquad \chi^2 = \sum_{j=1}^{M} \frac{\left[\hat{g}_1(\tau_j) - g_1(\tau_j)\right]^2}{\sigma_{g_1}^2(\tau_j)} \lesssim M. \tag{58}$$

Here, $g_1(\tau_j)$ denotes the measured field correlation function, $\hat{g}_1(\tau_j)$ its reconstructed counterpart, $\sigma_{g_1}(\tau_j)$ the experimental standard deviation at delay time τ_j, and M the total number of data points.

As in any measurement, before mathematical analysis can be meaningfully applied, the error—or at least an approximate error model—must be known [22]. This requirement is particularly important for MEM. The mathematical formulation of constrained MEM requires that the noise is independent and uncorrelated (homoscedastic). Only under these assumptions does the right-hand side of Eq. (58) behave as a true χ^2 statistic and therefore provide statistically meaningful weighting of residuals.

The central idea of entropy maximization is to expand small deviations in the exponential-decay domain into a much broader domain of distribution coefficients in order to recover the underlying distribution. This projection between domains is by design highly sensitive to small perturbations in the g_1 domain which puts a lot of weight on the noise.

Analysis of noise is laborious but necessary. In search of mathematical shortcuts, several analytical forms have been suggested [22]. The idea is to use the data itself, assuming that noise scales with the signal through some analytical relation. Two common forms are shown below in Eq. (59). Such models introduce statistical coupling between noise and data and therefore violate the statistical assumptions required for χ^2 weighting. In addition, as can be seen in Figure 9, both forms inevitably introduce additional structure into the χ^2 weighting and must therefore be treated with caution. Consequently, introducing an explicit noise model into χ^2 may bias the recovered distribution if the assumed noise shape significantly deviates from the actual experimental noise.

$$\sigma_j^2 = \frac{1 + d_j^2}{4Ad_j^2} \qquad \sigma_j^2 = \frac{1}{4A} d_j^2 \tag{59}$$

Here, A denotes an instrument constant related to photon counts, σ_j^2 are the error terms appearing in the denominator of the χ^2 function in Eq. (58), and d_j are the measured values of the autocorrelation function $g_1(\tau_j)$. In most commercial instruments the constant A is not directly accessible and must therefore be estimated.

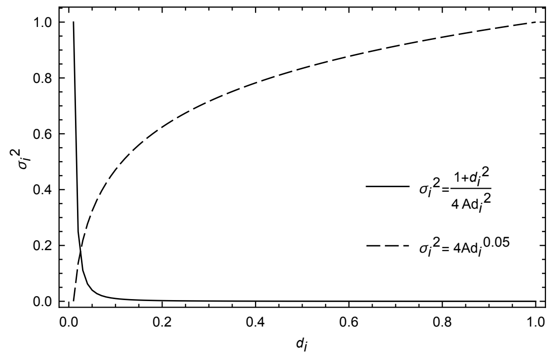

FIG. 9. Noise structure injected by analytical noise models into χ^2 statistic.

A seemingly simple way to avoid the entire noise-analysis problem is to use an unconstrained formulation of MEM. Instead of enforcing an experimentally estimated $\chi^2 = M$ to behave like the mathematically ideal one, a regularization parameter λ is introduced that controls how strongly entropy contributes to the fit. The optimization problem is then reformulated as the minimization of

$$\mathcal{Q}(\mathbf{a}) = \chi^2(\mathbf{a}) - \lambda S(\mathbf{a}), \tag{60}$$

where λ determines the balance between data and entropy contributions. However, the price for abandoning explicit noise analysis is that the choice of λ is no longer uniquely defined and must be guided by physical reasoning or heuristic criteria, such as the construction of the posterior distribution [26].

5. *Noise analysis*

Assuming that for rigid particles measured g_1-line shape reflects only polydispersity and instrumental noise, a reference sample with known polydispersity can be used to estimate the instrument noise contribution. For this purpose, a 60 nm polystyrene NIST (LTX3060A) standard was measured three times consecutively. This allows characterization of the temporal dependence and distribution of the noise, and guides the choice of data preprocessing and inversion strategy.

Figure 10 illustrates the noise structure as a function of delay time τ and reveals three distinct regions. In region I, the noise level is minimal and the signal is data-dominated. In region II,

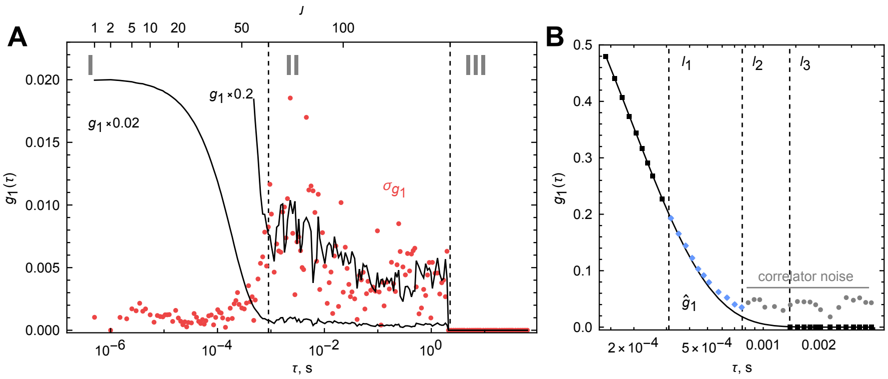

FIG. 10. Noise structure of the homodyne Malvern Nano S instrument exhibits three distinct regions: data-dominated (I), correlator-dominated (II), and hard cutoff (III). A — comparison of noise from three replicates σ_{g_1} (red points) with the corresponding g_1 function (black lines). B — preprocessing of the data guided by a single-exponential fit (blue line), including truncation (up to l_1 or l_2), removal (from l_1 or l_2 to l_3), and setting long-time data points to zero (13 points from l_3). Data points index j is shown on the top axis of A.

the signal is dominated by correlator noise. Region III corresponds to the sensitivity limit of the instrument, where the signal is effectively truncated to zero.

Meaningful physical information is contained primarily in region I. Therefore, data points that significantly deviate from the exponential decay should be truncated, as shown in Figure 10B. However, simply setting all long-time values of g_1 to zero is not appropriate, since assigning a value of zero is not equivalent to removing a data point and may distort the inversion. Instead, correlator-dominated data should be removed starting from the point where it departs from the exponential behavior. Only at delay times where the fitted $\hat{g}_1$ is effectively zero should the values be set to zero, with a small number of additional points retained to define a stable baseline for the MEM inversion.

6. χ^2 statistics applicability

As mentioned in section III C 3, the validity of the constraint in Eq. (58) depends on whether the residuals are uncorrelated and approximately normally distributed. From the three NIST replicates ($P = 3$), both the point residuals (numerator) and point errors (denominator) in Eq. (58) can be estimated in the following way. The experimentally measured correlation function $g_1(\tau)$ of each replicate is processed as illustrated in Figure 10B and then fitted by $\hat{g}_1(\tau) = \exp(-\tau/\tau_c)$, after which residuals are calculated as

$$r_k = \hat{g}_1(\tau_j) - g_1(\tau_j). \tag{61}$$

From the three replicates the standard deviation of the j-th residual is computed as

$$\sigma_{g_1}(\tau_j) = \sigma_j = \sqrt{\frac{1}{P-1}\sum_{k=1}^{P}(r_k - \bar{r})^2}. \tag{62}$$

These σ_j values are plotted in Figure 11A as red dots and labeled σ_{g_1} to avoid excessive notation.

A synthetic noise sample e_j of length M can be drawn from the normal distribution $\mathcal{N}(0, \sigma_n)$, where

$$\sigma_n = \sqrt{\frac{1}{M-1}\sum_{j=1}^{M}(\sigma_j - \bar{\sigma})^2}. \tag{63}$$

Here σ_n represents the standard deviation of the distribution of all σ_j values and not the three replicates of individual data points.

After these three quantities are computed (r_j, σ_j, and e_j), their distributions and independence can be examined by plotting standardized distributions and computing autocorrelation functions. The temporal dependence of r_j (red), σ_j (blue), and e_j (yellow), together with their standardized distributions and autocorrelation functions, are shown in Figure 11. It is immediately evident from Figure 11A,B that the noise is neither normally distributed nor independent (Figure 11C).

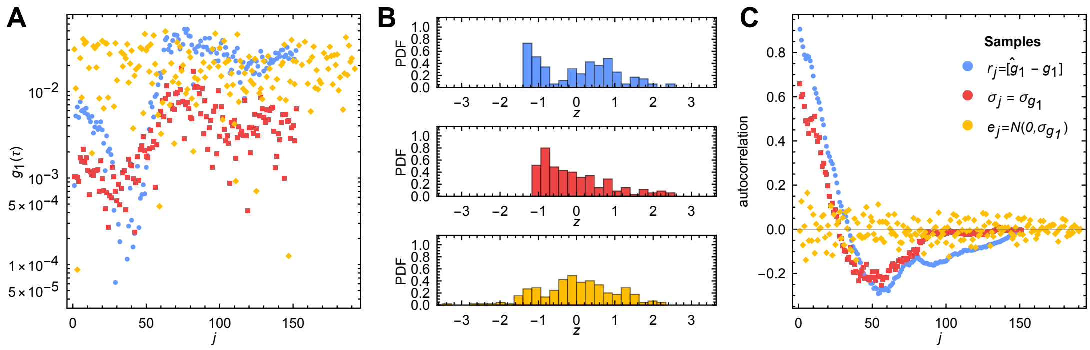

FIG. 11. Noise of the homodyne Malvern Nano S instrument is autocorrelated and non-Gaussian. A — squared residuals (blue) and squared experimental errors σ_{g_1} (red) exhibit a distinct temporal structure, in contrast to synthetic $\mathcal{N}(0, \sigma_{g_1})$ noise generated using the same σ_{g_1}. B — distributions of standardized residuals and errors deviate significantly from normality. C — residuals and experimental errors are strongly correlated, unlike the synthetic noise. PDF denotes probability density function, and j is the index of the g_1 data points. Sample is NIST polystyrene standard 60 nm.

Both of the conditions required for χ^2 to behave as a statistical measure are therefore violated. Consequently, it does not scale with M and cannot be used as a statistically meaningful constraint. For example, in the data-driven region I (Figure 10A) truncation to 50 points ($g_1 \approx 0.2$) results in $\chi^2/M = 7.4 \pm 2.4$. Adding 10 additional points shifts χ^2/M to 9.4 ± 2.0, both values being significantly larger than 1. Therefore, classical constrained MEM in the form of Eq. (58) is not valid for the data obtained with homodyne instrument.

7. *Geometrical feasibility boundary regularization. GFBR*

In this work a new approach is proposed. Instead of attempting to transform the data to comply with the statistical χ^2 constraint, a combination of geometrical and computational ideas is used:

1. Since noise increases as the magnitude of g_1 decreases, the g_1 function is truncated to the region where it remains close to the physically expected single-exponential behavior.

2. Instead of the statistical χ^2 constraint, the geometrical Euclidean distance (κ) is used as a misfit criterion.

3. The constraint feasibility boundary (C_b) is determined through the convergence behavior of the optimization algorithm within a fixed number of iterations.

4. The coefficient of variation (CV) of the short-time part of the recovered distribution (left side of the peak) is used to scan the region of constraints $C = C_b..2\text{–}3C_b$ with logarithmic steps in C, producing the curve $\mathrm{CV}(C)$. Mathematica function `dlsMEMc`.

5. The derivative $d \log \mathrm{CV}/d \log C$ is used to determine the geometric knee of this curve. This point corresponds to the transition between purely data-driven inversion and entropy-dominated inversion and therefore approximates the true distribution width.

6. Finally, the g_1 function truncated at the noise floor is analyzed using unconstrained MEM, and the regularization parameter λ is selected such that the resulting distribution width matches the knee estimate obtained from the constrained analysis. Mathematica function `dlsMEMu`.

7. For mathematical reasons related to the entropy optimization itself (see section III C 10), MEM systematically underestimates the distribution width. This requires the use of a reference sample with known distribution width in order to estimate the compression correction factor for each particular experimental setup. The final correction step is then to match the width obtained from unconstrained MEM to the corrected constrained width.

In this work the proposed approach is referred to as Geometrical noise boundary regularization (GFBR).

The constrained problem becomes

$$\kappa = \sum_{j=1}^{M} \left[\hat{g}_1(\tau_j) - g_1(\tau_j)\right]^2 \leq C, \qquad S(\mathbf{a}) = -\sum_{i=1}^{N} a_i \ln a_i, \qquad a_i \geq 0, \qquad \sum_{i=1}^{N} a_i = 1. \tag{64}$$

Here M is the number of data points in the measured correlation function g_1, and N is the number of points in the MEM kernel grid defined in Eq. (47). Conversely, the unconstrained formulation becomes

$$\mathcal{Q}(\mathbf{a}) = \kappa(\mathbf{a}) - \lambda S(\mathbf{a}), \tag{65}$$

The described approach addresses two problems simultaneously: the presence of correlated nonlinear noise in homodyne DLS data and the selection of the regularization parameter λ. The first issue is mitigated by using unconstrained MEM on the truncated correlation function. The second is resolved by matching the distribution width obtained from the unconstrained inversion to the geometrically determined knee of the constrained problem.

Exploiting the physical fact that the decay of $g_1(\tau)$ originates from exponential relaxation dynamics, truncation of the correlation function to the region where this behavior is preserved imposes a natural feasibility condition on the inversion. In this regime there exists a minimal constant C_b below which the constrained problem Eq. (64) has no admissible solution. Determining this constant therefore provides a physically grounded upper bound on the instrumental noise for each sample and allows the regularization parameter λ in Eq. (65) to be selected using purely geometric considerations.

A final remark should be made regarding the mathematical interpretation of the procedure. The CV–C curve used here plays a role analogous to the classical L-curve employed in regularization theory. However, in the proposed formulation both axes have direct physical interpretation: C represents the admissible level of deviation between the measured and reconstructed correlation functions, while CV characterizes the width of the recovered relaxation-time distribution. This makes the regularization space more transparent than the conventional (λ, S) representation used in entropy-based inversion methods.

8. *Data processing and MEM implementation*

The DLS instrument used in the present work is of homodyne type, and returns the second-order intensity autocorrelation function, $g_2(\tau) - 1$, which must be converted to the first-order field autocorrelation function, $g_1(\tau)$, before applying any analysis. In this work, the following procedure was used for this conversion.

The square root of the normalized $g_2(\tau) - 1$ was used to obtain $g_1(\tau)$:

$$g_1(\tau) = \sqrt{\frac{g_2(\tau) - 1}{\max[g_2(\tau) - 1]}}. \tag{66}$$

After conversion, the g_1 function was fitted with a single exponential of the form $B\exp(-bt)$, which was used as a guide line to both find the point where the correlator noise took over signal and the point where g_1 reached 0, lines l_1 and l_2, respectively. As shown in Figure 10B, the data points between the two vertical lines l_1 and l_2 were removed, and after the l_2 line, 13-20 points with $g_1 = 0$ were added (function `clipG1toG2`). This approach removes most of the correlator noise but at the same time does not introduce new information. Such truncation with replacement was applied for all samples prior any analysis.

The functions required for data processing and all operations described above were written in Mathematica 14.3 and collected in the file `"dls_mem.nb"`. The Interior Point OPTimizer (IPOPT) method was used for the optimization routines. A short description of the functions is provided inside the notebook. The data corresponding to the C-sweeps for the files discussed in the next section are stored as `.h5` datasets.

9. *Application of MEM to standard samples*

The final step in the whole DLS data analysis pipeline is the application of MEM inversion to standard samples, i.e. samples whose distribution is known or approximately known *a priori*, so that the accuracy of the inversion can be evaluated.

> Three samples were chosen for this analysis: the NIST 60 nm polystyrene standard (LTX3060A), apoferritin (AFt, Cytiva 28-4038-42, HMW, 440 kDa, 1.887(18) mg mL^{-1}), and bovine serum albumin (BSA, 05470-1G, 98%, CAS 9048-46-8, 2.508(12) mg mL^{-1}) solution. The NIST sample was diluted in MQ water. BSA and AFt were diluted in 50 mM HEPES A buffer (**??**). BSA was filtered through a Whatman 0.02 µm syringe filter, whereas AFt was not filtered because of its larger size. Samples were measured in Sarstedt 67.758 cuvettes using a sample volume of 50 µL on a Malvern Nano S instrument at 23 °C, with laser wavelength $\lambda = 633$ nm and scattering angle $\theta = 173°$.

The application of the GFBR framework to the three samples is shown in Figure 12. The extracted distribution positions are physically reasonable (Table V), demonstrating that the methodology can be successfully applied to unknown samples. The NIST sample (CV 13.4%) illustrates that the recovered distribution width was underestimated by approximately 17%. Nevertheless, the radius calculated from the peak position was approximately 30 nm, in close agreement with the nominal particle radius. BSA and AFt radii are also in good agreement with literature values. This physically grounded correction factor was then used to correct the unconstrained MEM distribution width in step 7 of the GFBR routine.

The amplitudes a_i should not be interpreted as independent fit parameters, but rather as a conserved probability mass redistributed across logarithmic bins. The geometrical interpretation of MEM can be visualized in the following way. The recovered distribution behaves similarly to an air balloon filling a mold. However, neither the form nor the balloon are isotropic, and therefore the match between them depends on the applied pressure. At sufficiently low pressure, the balloon only partially conforms to the shape of the form, corresponding to underfitting. At excessively high pressure, the balloon deforms beyond the physically meaningful geometry and effectively escapes through the entrance of the form, corresponding to overfitting. At some intermediate pressure, however, the mismatch between the balloon geometry and the molding form becomes minimal. Determining this optimal pressure is the central difficulty of MEM optimization. In general, the solution is not unique, and one possible approach for estimating it is the GFBR framework presented above. In this analogy, the regularization parameters C or λ control the effective pressure applied to the balloon.

MEM employs logarithmic discretization, which is both physically and numerically motivated. Relaxation processes in DLS commonly span several orders of magnitude, while the Laplace transform kernel depends exponentially on $\gamma\tau$. Uniform discretization in linear γ space therefore allocates most grid points to rapidly decaying components and under-samples slow processes. Logarithmic spacing produces approximately scale-invariant resolution across the entire dynamic range and significantly improves numerical stability of the inversion. Consequently, after performing MEM, the recovered intensity per logarithmic bin must be converted to intensity per linear bin.

On a logarithmically spaced grid in γ, the amplitudes a_i represent integrated probability over logarithmic intervals $\delta\log(\gamma)$ rather than pointwise probability density in linear γ space. Consequently, direct visualization of a_i systematically overrepresents long relaxation times because logarithmic bins become progressively wider in linear coordinates. This transformation is equivalent to the standard Jacobian correction between logarithmic and linear coordinates.

To recover the corresponding probability density function (PDF), the amplitudes must therefore be divided by the logarithmic bin width,

$$p(\gamma_i) = \frac{a_i}{\delta\log(\gamma_i)}. \tag{67}$$

The same transformation applies in the relaxation-time representation,

$$p(\tau_i) = \frac{a_i}{\delta\log(\tau_i)}, \tag{68}$$

since $\tau = 1/\gamma$ preserves logarithmic spacing.

It should be noted that, from a practical perspective, both intensity-weighted and density-weighted distributions are useful. The PDF transformation strongly suppresses long-time components because the logarithmic bins become progressively wider at large τ. Consequently, if the distribution contains both short-time and long-time components, peak positions are often more accurately determined from the intensity-weighted representation, while distribution shapes are more reliably analyzed using the PDF representation. For example, for BSA and AFt, the distributions at C_b show the presence of two peaks whose positions can be determined reasonably accurately (Figure 12C,D). Therefore, feasibility-boundary scans together with both

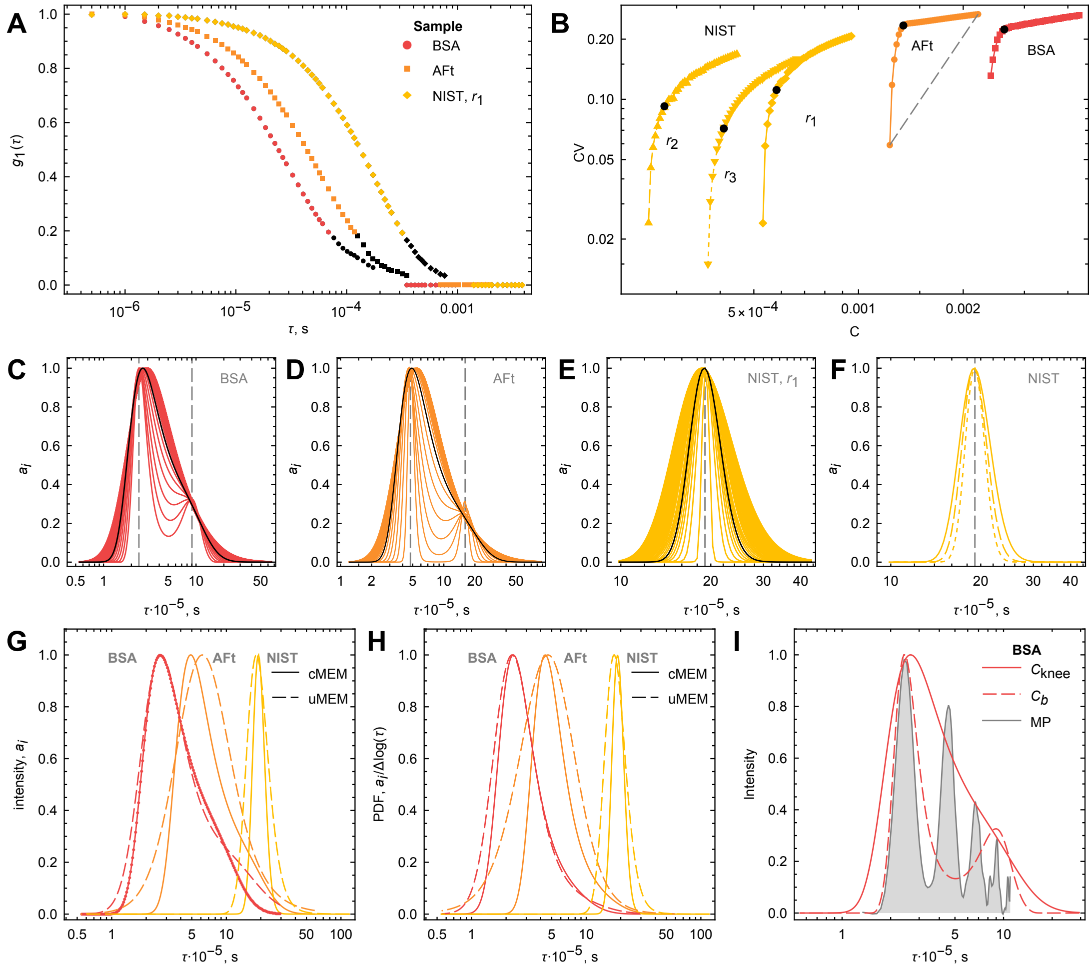

FIG. 12. GFBR MEM inversion of three standard samples illustrates the principle of the method. A — g_1 functions processed as described in section III C 5. Black points illustrate the difference between truncation used for constrained and unconstrained MEM. B — CV–C curves and geometrical determination of the knee points, shown as black dots. C, D, E — scans from the feasibility boundary region. The narrowest distribution corresponds to the feasibility boundary C_b in step 3. Black curves correspond to the knee points shown in panel B. F — three replicates of the NIST standard sample at the corresponding knee points from panel B. G — intensity distributions obtained from constrained (solid) and unconstrained (dashed) MEM. The BSA curve contains points illustrating the logarithmic grid spacing. H — probability density distributions $\frac{a_i}{\delta \log(\tau)}$ obtained from constrained (solid) and unconstrained (dashed) MEM. I — Comparison of MEM and mass photometry distributions. The shaded area represents the MP distribution converted into the relaxation-time domain. The agreement is surprisingly good: MEM at C_b reproduces the positions of the monomer and tetramer peaks, indicating that the method is capable of resolving the dominant hydrodynamic modes. The correspondence between the MP aggregation-number ratio (3.64 ± 0.17) and the MEM relaxation-time ratio (3.74) suggests an approximately linear scaling $R_h \propto N$ between aggregation number and hydrodynamic size for small BSA oligomers, in contrast to the compact-sphere scaling $R_h \propto N^{1/3}$.

intensity-weighted and PDF representations provide substantially more information than the final optimized PDF alone, and should be used as an exploratory research tool rather than solely as a final inversion procedure.

TABLE V. MEM geometrical regularization parameters for standard samples. λ_{knee} corresponds to the corrected value obtained in step 7. Hydrodynamic radii R_c and R_u were calculated from τ_c and τ_u using the Einstein–Stokes relation.

	AFt	BSA	NIST r_1	NIST r_2	NIST r_3
C_b	0.0012	0.0024	0.00053	0.00025	0.00037
C_{knee}	0.0013	0.0026	0.00058	0.00028	0.00041
λ_{knee}	0.042	0.012	0.0014	0.00039	0.00023
τ_c (10^{-5} s)	4.38	2.32	18.74	18.74	19.12
R_c (nm)	7.5	4.0	31.9	31.9	32.6
τ_u (10^{-5} s)	4.65	2.27	17.65	17.65	18.0
R_u (nm)	7.9	3.9	30.1	30.1	30.7

10. Entropy limitations in inversion of g_1 from power-law samples

As shown by the application of the GFBR MEM routine to standard samples, the distribution widths recovered at the knee point C_{knee} are underestimated by approximately 17%. The origin of this effect is that entropy maximization constrains distribution broadening only logarithmically. For example, for a Gaussian distribution the differential entropy scales as

$$H \propto \ln \sigma. \tag{69}$$

Consequently, entropy grows only logarithmically with increasing distribution width. Numerically, the natural logarithm increases by only ≈ 2.3 per decade. Thus, doubling the entropy requires increasing the standard deviation from $\sigma = 0.5$ to $\sigma = 5$, while an additional doubling requires another tenfold increase from $\sigma = 5$ to $\sigma = 50$, see Figure 13B. In contrast, the misfit term grows quadratically with deviations from the experimental data. Therefore, the optimization is intrinsically asymmetric with respect to entropy and misfit contributions, producing an inherent preference for narrower distributions.

While the power-law example in Eq. (6) is defined analytically, the simulations in section III A illustrating the sampling problem were performed on a finite interval of sizes 1–100, since any aggregating or evolving system cannot grow indefinitely and particles above some critical size are therefore absent. This finite-size truncation is a well-established concept in coagulation theory [28, 32, 33].

For a truncated power-law distribution

$$p(z) = Az^{-\alpha}, \qquad z \in [z_0, b], \tag{70}$$

where the normalization constant A is determined by

$$\int_{z_0}^{b} p(z)\, dz = 1, \tag{71}$$

the Shannon entropy is given by

$$S = -\int_{z_0}^{b} p(z) \log p(z)\, dz = -\int_{z_0}^{b} Az^{-\alpha} \log\left(Az^{-\alpha}\right)\, dz. \tag{72}$$

For fixed values of the normalization constant A, lower cutoff z_0, upper cutoff b, and scaling exponent α, the entropy S is constant. Consequently, unlike localized distributions such as the Gaussian family, a truncated self-similar distribution does not possess a continuous deformation parameter analogous to the standard deviation. This observation suggests that the limitation of MEM is deeper than weak entropy sensitivity alone. For truncated scale-free distributions, the entropy functional itself contains compensating scaling contributions arising from the simultaneous decrease of probability density $p(z)$ and the logarithmic weighting term in the entropy integrand Eq. (72). Consequently, after normalization on bounded support, optimization of entropy no longer corresponds to optimization of physically meaningful deformation of the distribution geometry, see Figure 13C.

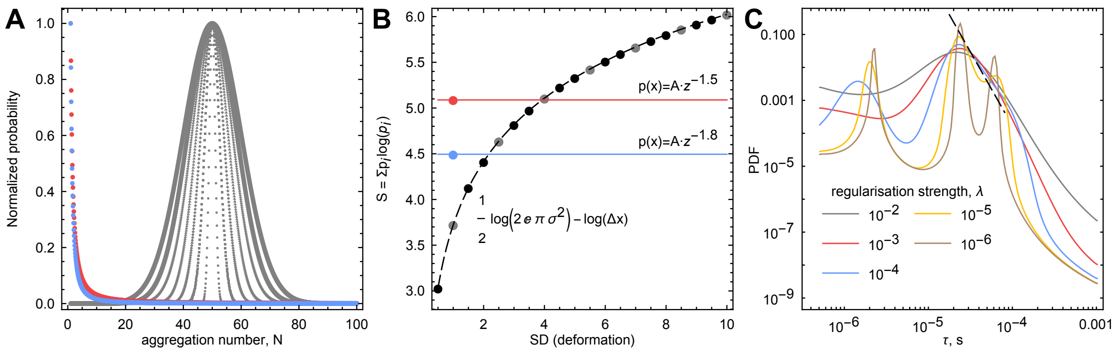

FIG. 13. Generation of artifacts via MEM in scale-free systems. A — Synthetic power-law distributions (red: $N^{-1.5}$, blue: $N^{-1.8}$) and Gaussian distributions (gray) $\mathcal{N}(50, \sigma)$ with $\sigma = 0.5 \ldots 10$ in steps of 0.5, generated on a fixed support $N = 1 \ldots 100$ with discretization step 0.1. The distributions were discretized on the same finite interval so that the discrete Shannon entropy could be calculated consistently with the MEM methodology. B — Entropy as a function of the Gaussian deformation parameter σ. Black points span the full range of simulated Gaussian widths, while gray points correspond to the Gaussian distributions shown in panel A. Red and blue lines correspond to the entropies of the power-law distributions in panel A. Since the power-law distributions lack a continuous deformation parameter analogous to σ, their entropy remains invariant. C — MEM inversion of power law distribution (dashed line) using the unconstrained `dlsMEMu` implementation over a wide range of regularization parameters (colored solid lines). The recovered distributions exhibit clearly defined artificial peaks, demonstrating compression of the underlying scale-free distribution into localized effective structures as a consequence of entropy regularization.

Instead, the physically relevant quantities become the scaling variables themselves, such as the characteristic relaxation time τ, critical size, or dynamic scaling factor. Therefore, scaling-based approaches are mathematically more robust for self-similar systems, because they preserve the underlying invariance of the distribution rather than attempting to reconstruct it through entropy deformation.

For DLS, the scaling argument allows direct use of the autocorrelation function $g_1(\tau)$. Since the underlying distribution is assumed to be self-similar and scale-free, the problem reduces to determining the mapping between characteristic relaxation times and aggregate size. In the present work, the approach developed by Martin [28] is used.

The artifacts generated by applying MEM to power-law distributions illustrate an important fundamental limitation: the distribution class cannot be reliably inferred from measurements of $g_1(\tau)$ alone. Additional orthogonal experiments or strong theoretical assumptions are therefore required. Although this limitation is demonstrated here for DLS, the same inverse-problem ambiguity is endemic to many relaxation-based techniques, including FCS, NMR, fluorescence lifetime spectroscopy, and related methods. Consequently, appropriate mathematical formalism should not be selected solely from the measured relaxation signal, but must instead be constrained by independent experimental or theoretical information.

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